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Journal of Non-Crystalline Solids 345&346 (2004) 70–74
JOURNAL OF NON-CRYSTALLINE SOLIDS
www.elsevier.com/locate/jnoncrysol

NMR and Mössbauer characterization of tin(II)–tin(IV)–sodium phosphate glasses

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Available online 6 October 2004

Abstract

xSnO–(100 − x)$NaPO_3$ glasses were prepared under argon atmosphere for $0 \leqslant x \leqslant 40$. Wet chemical analyses and Mössbauer spectroscopy indicate the presence of Sn(II) and Sn(IV) in the glasses, with a majority of Sn(II). ^{119}Sn NMR and Mössbauer spectra show that the local environment of Sn(II) are characteristic of SnO in network forming sites. The ^{119}Sn NMR spectrum of an oxidized glass shows that Sn(IV) are located in sites similar to SnP_2O_7. ^{31}P MAS–NMR spectra show that the sodium metaphosphate chains are depolymerized by SnO, and that dissociation into the phosphate melt is complete. The ^{31}P resonances were assigned to middle-chain group charge compensated by Na^+(Q^2(Na)), chain-end group charge compensated by Sn^{2+}(Q^1(Sn(II))) and Na^+(Q^1(Na)). The ^{31}P MAS–NMR spectrum of the oxidized glass enabled to measure the chemical shift of (Q^1(Sn(IV))), and the double-quantum filtered ^{31}P MAS–NMR spectrum showed that Sn(IV) sites are inserted in the phosphate glass network.

PACS: 61.43.F; 61.18F

1. Introduction

Phosphate glasses formulated with SnO were developed as alternative lead-free low temperature sealing glasses [1], and cathode materials for lithium batteries [2]. In these glasses, only Sn(II) contributes positively to the properties. Indeed, Sn(IV) has a low solubility in glasses, and it increases the glass transition temperature. We recently proposed a new method for the immobilization of heavy metal chlorides ($PbCl_2$, $CdCl_2$) that may volatilize during the fly ash vitrification process of [3]. The method is based on the conversion of the chlorides into non-volatile phosphates. Tin chloride, despite its volatility at low temperature (650 °C), can also be converted into a phosphate salt, which can be vitrified without losses. This process produces sodium–tin phosphate glasses, which compositions that can be simplified as xSnO–(100 − x)$NaPO_3$. The present paper reports a study of the structure and properties of this glass system, with a particular attention paid to the effect of Sn(IV), which may be formed during the conversion process. The glasses were first prepared under inert atmosphere and the glass redox (R = Sn(II)/Sn_{total}) was measured. Then, a particular composition was oxidized in order to produce Sn(IV).

2. Experimental procedures

xSnO–(100 − x)$NaPO_3$ glasses were prepared from batches of reagent grade SnO and $NaPO_3$. Each batch

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0022-3093/$ - see front matter
doi:10.1016/j.jnoncrysol.2004.07.070

was melted in a carbon crucible at 1000 °C for 30 min, under argon. The glass with the smallest Sn amount (5%) was oxidized under air at 1000 °C for 1 h. The glasses were analyzed by inductively coupled plasma atomic emission spectrometry (ICP-AES). The Sn(II) concentration was measured by polarography in order to calculate the redox ($R = Sn(II)/Sn_{total}$). The redox was also checked with tin Mössbauer on some glass compositions, and it was found to be in agreement with measurements obtained in solution.

The Mössbauer spectra were obtained using a constant acceleration spectrometer with a room temperature $CaSnO_3$ source (119Sn) in a transmission geometry. The samples contained 15 mg of natural tin per square centimeter. For such a concentration, line broadening due thickness effects can be neglected. The spectra at 4.2 K and 293 K were recorded using a variable temperature cryostat. The spectra were fitted as a sum of Lorentzians by least square refinements. All isomer shifts given in this paper refer to $CaSnO_3$ at 293 K.

The ^{31}P MAS–NMR spectra were recorded at 2.34 T, with a 1.5 μs pulse (45°), and a 60 s delay between each accumulation to enable full relaxation. A 7 mm MAS probe was used. The MAS–NMR spectra were decomposed with the DM-FIT software [4]. Gaussian lines were used since ^{31}P resonances in glasses are dominated by chemical shift distribution. However, a small Lorentzian contribution (ca. 5%) was used to optimize the fits. The ^{31}P MAS–NMR double-quantum spectrum was recorded at 2.34 T. The spinning frequency was set to 5 kHz, leading to time increments in the t_1 dimension (DQ) of 200 μs to record the rotor synchronized two dimension spectrum. The back-to-back acquisition sequence has been used, TPPI method for phase incrementation. Sixty four pulses of 3 μs (90°) were used for double-quantum coherence excitation and reconversion. The repetition time was 60 s (pre-saturation was used) and 64 increments were necessary in the t_1 dimension to record the whole signal. On DQ spectra, the resonances are labeled by their chemical shifts on the MAS and double-quantum axes ($\delta_{MAS}, \delta_{DQMAS}$). The ^{31}P NMR chemical shifts are referenced to an 85% H_3PO_4 solution at 0 ppm. The $Q^{n,ij}_{(M)}$ notation is used: n is the number of bridging oxygens bonded to a given PO_4 site, i and j indicate the nature of the other Q^n sites bonded through these bridging oxygens, and (M) indicates which cation is compensating the charge of the PO_4 site.

The ^{119}Sn NMR spectra were recorded at 9.4 T (Larmor frequency 105.8 MHz), with 2 μs pulse (45°) and 60 s delay between each accumulation. A 4 mm probe was used, but the spectra were recorded in static condition since the large CSA of tin, due to a large electronic shielding, cannot be removed by MAS. ^{119}Sn NMR chemical shifts are referenced to 0.1 M aqueous $SnCl_2$.

3. Results

Table 1 shows the batch and analyzed glass compositions. The total Sn quantity (Sn(II) + Sn(IV)) is expressed as SnO_{tot}. The redox of the glasses (Sn(II)/(Sn(II) + Sn(IV)) is always high (between 0.95 and 0.99) since the glasses were prepared under argon atmosphere. In the glass oxidized under air, all Sn(II) were transformed into Sn(IV), leading to a redox value of 0 (Table 1). Oxidation could not be realized on the other glasses since opacities occurred, which are due to crystal precipitation as will be discussed below.

The Mössbauer spectra (Fig. 1) can be decomposed into three peaks: a quadrupolar doublet with isomer shift ($3.11 \leqslant \delta \leqslant 3.30\,\mathrm{mm\,s^{-1}}$) characteristic of Sn(II), and a singlet corresponding to Sn(IV) ($-0.45 \leqslant \delta \leqslant -0.35\,\mathrm{mm\,s^{-1}}$) [5]. The Sn(IV) signal is weak, in accordance with its low concentration in the glasses. The quadrupolar splitting of Sn(IV) is very small [5], it appears as a singlet. Thus, the asymmetry of the Sn(II) doublet is not due to overlapping with a Sn(IV) line as suggested in Ref. [6]. At 293 K, the asymmetry of the Sn(II) doublet can arise from two different effects: an influence of the texture or an anisotropy of the Debye–Waller factor (Goldanskii–Karyagin effect). The Mössbauer spectrum recorded at 4.2 K (Fig. 2) is symmetric. This confirms the Goldanskii effect, due to the vibrational anisotropy associated with the presence of the $5s^2$ lone pair of Sn(II). The asymmetry on 293 K spectra remains as the quantity of SnO increases, which means that the coordination sphere of Sn(II) is barely affected by the composition.

^{119}Sn NMR spectra are shown in Fig. 2. For the glasses prepared under neutral atmosphere, the broad spectra are characteristic of Sn(II). Indeed, the large electronic shielding of the Sn(II) induces a large chemical shift anisotropy that cannot be suppressed by Magic Angle Spinning. The ^{119}Sn spectra are identical whatever the SnO quantity in the glasses xSnO–$(1-x)NaPO_3$, contrary to the spectra of ySnO–$(100-y)P_2O_5$ glasses ($y = 70$ and $y = 55$ in Fig. 2). A shift of Sn(II) resonance, also reported in [7], is observed for the latter when y increases from 55 to 70. The spectrum of the oxidized glass ($x = 5$ oxidized in Fig. 2)

Table 1
Batch and analyzed compositions of xSnO–$(100-x)NaPO_3$ glasses, and redox values

SnO_{tot} batch (mol%)	$NaPO_3$ batch (mol%)	SnO_{tot} analyzed (mol%) ± 2	Na_2O analyzed (mol%) ± 2	P_2O_5 analyzed (mol%) ± 2	Redox ±0.02
40	60	39.5	30.6	29.9	0.95
30	70	28.5	35.9	35.6	0.99
15	85	15.7	39.8	44.5	0.96
5	95	5.3	45.0	49.7	0.98
5	95	5.2	45	94.8	0

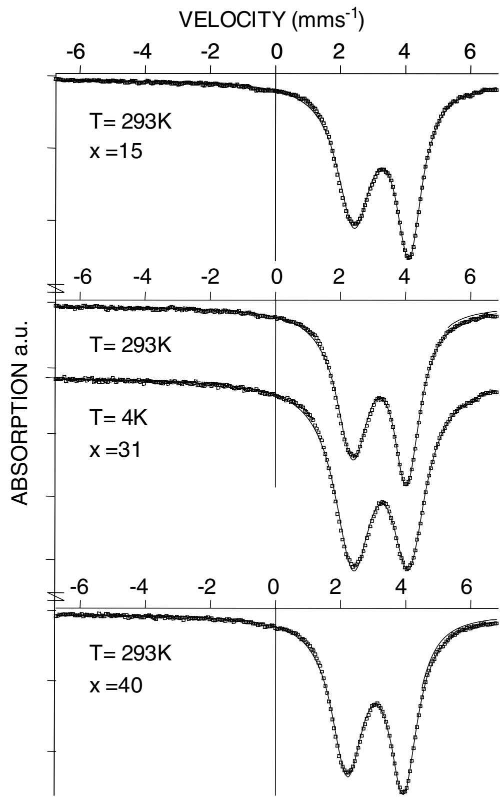

Fig. 1. Mössbauer spectra of xSnO–(100 − x)$NaPO_3$ glasses.

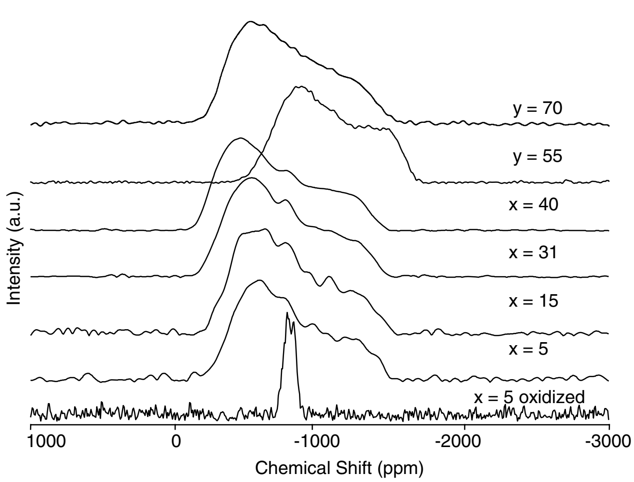

Fig. 2. ^{119}Sn NMR spectra (static) of xSnO–(100 − x)$NaPO_3$ glasses and ySnO–(100 − y)P_2O_5.

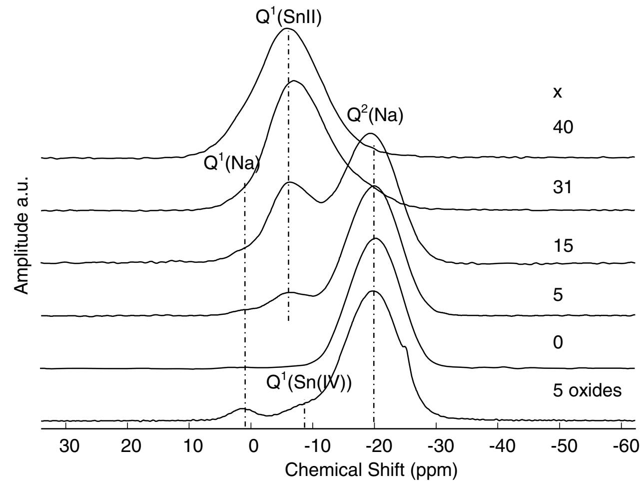

Fig. 3. ^{31}P MAS–NMR spectra of xSnO–(100 − x)$NaPO_3$ glasses.

shows a narrow resonance due to Sn(IV), with a chemical shift anisotropy that is much smaller than for Sn(II). This CSA is similar to that observed in SnO_2 or SnP_2O_7 crystals for which the coordination sphere is octahedral. This suggests that Sn(IV) environment in xSnO–(100 − x)$NaPO_3$ glasses is also very symmetric, as was also reported in binary xSnO–(100 − x)P_2O_5 [7]. Moreover, the isotropic chemical shift of Sn(IV) resonance (−820 ppm) is closer to that observed in SnP_2O_7 crystal (−850 ppm) than that observed in SnO_2(−603 ppm). This leads to the conclusion that Sn(IV) are bonded to Q^1 sites, as will be shown below. Due to the small quantity of Sn(IV) in the non-oxidized glasses (x = 5–40 in Fig. 2), Sn(IV) appears as a weak shoulder, at the same chemical shift than in the oxidized glass.

The ^{31}P MAS–NMR spectrum of $NaPO_3$ (Fig. 3) shows a single resonance for Q^2 sites bonded to Na^+ at −20 ppm (Q^2(Na)) [8]. The introduction of SnO results in the formation of Q^1 sites at −6 ppm, mainly bonded to Sn(II) (Q^1(Sn(II))), because the influence of Sn(IV) can be considered as negligible. Indeed, the spectrum decomposition with two Gaussian lines (not shown) gives a Q^1/Q^2 ratio of 6.5/93.5, close to the value calculated from the glass composition: 5/95. However, the bonding of these Q^1 sites to Sn(II) does not exclude a participation of Na^+, since the chemical shift of Q^1(Sn(II)) in $Sn_2P_2O_7$ crystal is more shielded (two sites at −12 and −15 ppm) than in the glasses (−6 ppm). A very weak resonance, at 2 ppm, is attributed to Q^1 sites mainly bonded to Na^+(Q^1(Na)), since its chemical shift is close to that observed in binary sodium phosphate glasses [8]. The increase of SnO quantity in xSnO–(100 − x)$NaPO_3$ glasses results in an increase of Q^1(Sn) and Q^1(Na) sites at the expense of Q^2(Na), meaning that SnO is integrated into the phosphate network through POSn bonds.

After the oxidation treatment, the ^{31}P MAS–NMR spectrum (Fig. 3) still shows Q^2(Na) and Q^1(Na) sites (Fig. 3). Since only Sn(IV) is present in the oxidized glass (Table 1), the resonance at 8 ppm can be unambiguously attributed to Q^1(Sn(IV)). As for Q^1(Sn(II)), some mixed bonding of Q^1(Sn(IV)) must occur with Na^+ since the chemical shift of pure Q^1(Sn(IV)) in SnP_2O_7 crystal is spread between 20 and 40 ppm [9,7]. A small narrow resonance a visible on the high field side of the spectrum of the oxidized glass, at 25 ppm. Although the visual

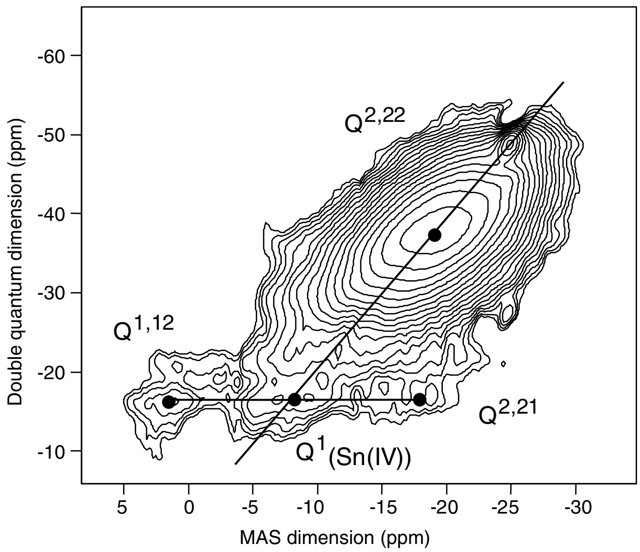

Fig. 4. ^{31}P Double-quantum filtered MAS–NMR spectrum of the 5SnO–95NaPO$_3$ oxidized glass.

inspection did not reveal opacity, the chemical shift is the same as reported for precipitated crystals from ySnO–$(100-y)$P$_2$O$_5$ glasses [7]. Those crystals were identified with X-ray diffraction as SnP$_2$O$_7$ [7]. Thus, NMR indicates that SnP$_2$O$_7$ is actually present in this glass. But its quantity is less than 1% of the NMR spectrum, and it could not be detected by X-rays diffraction.

In order to obtain more information about the connectivity of the Q^1 sites, we recorded the ^{31}P double-quantum filtered MAS–NMR spectrum of the oxidized glass (Fig. 4). This method enables to evaluate the spatial proximity of Q^n sites. During the evolution of magnetization under double-quantum coherence, the dipolar interaction, averaged out under MAS conditions, is reintroduced. Since the dipolar interaction is proportional to r^{-3} (r is the internuclear distance), the double-quantum filtered signal contains distance information. On the two-dimensional double-quantum spectra, signals lying on diagonal are self-correlated like for instance (Q^1 sites of pyrophosphates (diphosphates)), and off-diagonal signals indicate connectivity between sites of different environments (Q^1 sites of chain-end groups correlated with Q^2 sites). The ^{31}P DQ MAS–NMR spectrum of the 5SnO–95NaPO$_3$ oxidized glass, presented on Fig. 4, contains a main self-correlated resonance at (−19, −38) ppm, due to Q^2 sites connected to two other Q^2 sites. According to Witter coworkers [10], they are named $Q^{2,22}$. The Q^1(Na) sites at 2 ppm, located off-diagonal on the DQ spectrum (2, −16) ppm, are also correlated to another off diagonal resonance at (−17, −16) ppm. These correlations are typical of Q^1 sites of chain-end group ($Q^{1,2}$), connected to Q^2($Q^{2,21}$). The Q^1(Na) site at 2 ppm may have been assigned to Q^0 sites since it is deshielded compared to the Q^1(Sn(II)). However the double-quantum spectrum clearly indicates a correlation to Q^2, which excludes Q^0 sites. The Q^1(Sn(IV)) sites are located on the diagonal of the DQ spectrum (−8, −16) ppm, indicating they are that self-correlated. This means that those Q^1(Sn(IV)) occur in diphosphate units. The weak resonance at 25 ppm appears on the diagonal (self-correlated) of the DQ spectrum (−25, −50) ppm, which confirms its assignment to diphosphates SnP$_2$O$_7$ crystals.

4. Discussion

^{31}P MAS–NMR spectra give evidence for a dissociation of SnO into NaPO$_3$ with the formation of SnOP bonds, leading to Q^1(Sn) sites. The measured Q^1/Q^2 ratio is in good accordance with the calculated value, probably because the amount of Sn(IV) was limited in the glasses by the melting under inert atmosphere. The chemical shifts of the Q^n sites are barely modified by changing the SnO quantity, which means that the local environments of phosphate groups remain unchanged. This is in accordance with the ^{119}Sn NMR spectra: the chemical shift anisotropy and the center of gravity of the static spectra are not modified when the SnO quantity increases in the glasses (Fig. 2). As reported by Holland et al. [7], ^{119}Sn NMR spectra are typical of the presence of the 5s^2 electronic lone pair in the first coordination sphere of Sn(II) atoms, with a trigonal pyramid geometry. We confirmed the presence of this electronic lone pair with the Goldaanskii-Karyagin effect on Mössbauer spectra. However, contrary to what is observed for binary ySnO–$(100-y)$P$_2$O$_5$ glasses [7], in pseudo-binary xSnO–$(100-x)$NaPO$_3$ glasses the local Sn(II) environments are not modified with the quantity of SnO (Fig. 2). This is probably because the presence of Na$^+$ ions enables tin atoms to adopt a single coordination sphere. The CSA of ^{119}Sn NMR spectra, compared to these of ySnO–$(100-y)$P$_2$O$_5$ glasses (Fig. 2), clearly indicate that Sn(II) coordination is characteristic of network forming sites. In binary glasses ySnO–$(100-y)$P$_2$O$_5$, tin must act as both network modifier and former. Indeed, in this last case, the covalence of Sn–O bonds must change with the composition to take on this double role. On account of the large tin concentration in these glasses, and the large ionic radius of Sn(II) (0.093 nm), it would be of great interest to measure interatomic distances using X-rays or neutrons scattering.

The solubility of Sn(IV) is known to be very low in phosphate as well as in silicate glasses [1,11]. Holland et al. [7] proposed that in ySnO–$(100-y)$P$_2$O$_5$ glasses, Sn(IV) are located into SnP$_2$O$_7$ crystals which separate from the melt. However, we had no evidence for such phase separation in xSnO–$(100-x)$NaPO$_3$ glasses. This is probably because the melt basicity is higher due to the presence of Na_2O. Indeed, it increases the dissociation of SnP_2O_7 or SnO_2 crystals and thus increases the Sn(IV) solubility in the glasses. Moreover, the ^{31}P DQ NMR spectrum clearly shows that at least part of Sn(IV) are bonded to diphosphate sites which have a different chemical shift than SnP_2O_7 crystals. This deshielded chemical shift indicates that some Na^+ may be bonded to the diphosphate units together with Sn(IV). The broad line shape of their resonance on the DQ spectrum confirms that these diphosphate units are not located in crystallized phases, but are inserted into the glass network. Unfortunately, in the non-oxidized glasses the overlapping of the resonances of the different Q^n units does not enable to record a resolved DQ NMR spectrum. The association of Ti(IV) with diphosphates units, similar to Sn(IV) was also characterized by DQ NMR in other phosphate glasses [12].

5. Conclusion

Mössbauer and wet chemical analyses indicate that both Sn(II) and Sn(IV) are present in xSnO–$(100 - x)$NaPO$_3$ glasses. Mössbauer and ^{119}Sn NMR show that Sn(II) local environment is influenced by the $5s^2$ electronic lone pair. Contrary to the situation in ySnO–$(100 - y)$P$_2$O$_5$ glasses [7], the Sn(II) and Q^n environments in xSnO–$(100 - x)$NaPO$_3$ glasses are not modified by the quantity of SnO. Sn(II) are located in network forming site whatever the glass composition. ^{31}P double-quantum filtered MAS–NMR showed that Sn(IV) are bonded to diphosphate units inserted into the glass network, which are different from SnP_2O_7 phase separated crystals.

Acknowledgment

The authors are grateful to B. Revel (Centre Commun de Mesure RMN de l'USTL) who recorded some of the NMR spectra. E.B. acknowledges the ADEME and Région Nord Pas-de-Calais for providing her thesis grant.

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